

CS412 HW4

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1 Introduction

In this experiment, we apply fine-tuning techniques on a pre-trained VGG-16 model to classify images into two categories. The task involves training the model using different strategies and analyzing performance through training, validation, and testing phases. The objective is to identify the most effective fine-tuning strategy and learning rate.

2 Methodology

We fine-tuned the VGG-16 model using two different strategies:

- **Freeze all convolutional layers and train only the classifier head.**
- **Unfreeze the last convolutional block and the classifier head, and train them.**

We used the following hyperparameters:

- **Optimizer:** Adam
- **Loss function:** nn.BCEWithLogitsLoss
- **Epochs:** 10
- **Batch size:** 64
- **Learning rates:** 0.001 and 0.0001

3 Training and Validation Results

During training, the model's performance was evaluated using both training loss/accuracy and validation loss/accuracy after each epoch. The following table summarizes the key results:

Strategy	Learning Rate	Final Train Acc.	Final Val Acc.
Frozen Convs + Classifier Only	0.001	98.46%	95.70%
Frozen Convs + Classifier Only	0.0001	99.73%	96.27%
Unfreeze Last Block + Classifier	0.001	99.04%	97.77%
Unfreeze Last Block + Classifier	0.0001	99.95%	97.43%

Table 1: Performance comparison of different fine-tuning strategies and learning rates.

Observations:

- **Frozen Convs + Classifier Only (Learning Rate = 0.001):** The model initially shows good performance, but slight overfitting is observed towards the later epochs, with a final validation accuracy of 95.70%.
- **Frozen Convs + Classifier Only (Learning Rate = 0.0001):** The model achieves the highest stability, with a final validation accuracy of 96.27%. This setup performs best when keeping the convolutional layers frozen.
- **Unfreeze Last Block + Classifier (Learning Rate = 0.001):** The model’s performance significantly improves, achieving a validation accuracy of **97.77%**, the highest for this strategy. This suggests that fine-tuning the last block has a positive impact.
- **Unfreeze Last Block + Classifier (Learning Rate = 0.0001):** The model achieves a very high training accuracy of **99.95%**, but overfitting starts to appear after epoch 5. The validation accuracy remains strong at 97.43%.

4 Loss and Accuracy Plots

Below are the plots illustrating the training and validation loss as well as the accuracy over the 10 epochs for each configuration.

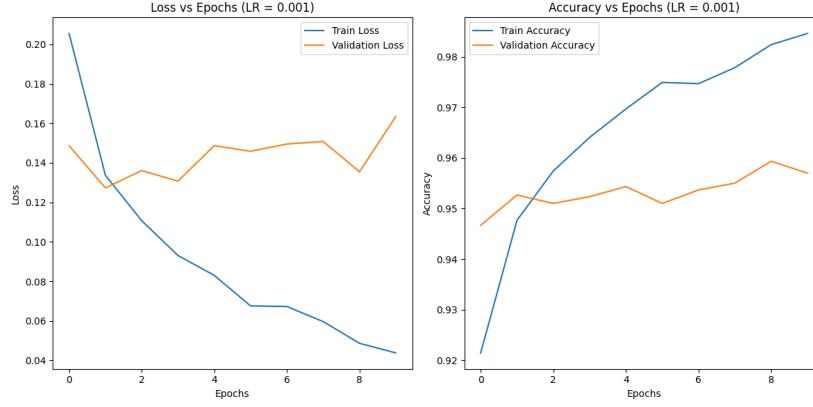


Figure 1: Loss and Accuracy vs Epochs for Learning Rate = 0.001 (Frozen Convs + Classifier Only)

Observations for Figure 1:

- **Loss and Accuracy Plot (LR = 0.001):** The training accuracy increases steadily until it stabilizes, but a slight dip in the validation accuracy towards the last epochs indicates overfitting. The validation loss increases, reflecting this overfitting.

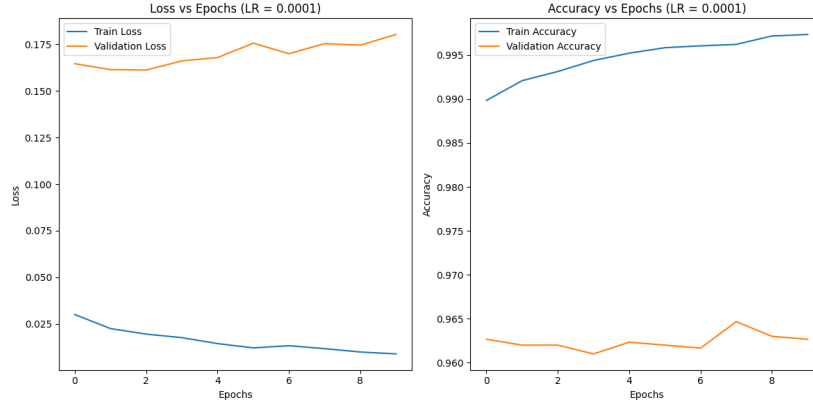


Figure 2: Loss and Accuracy vs Epochs for Learning Rate = 0.0001 (Frozen Convs + Classifier Only)

Observations for Figure 2:

- **Loss and Accuracy Plot (LR = 0.0001):** The model shows high stability, with both the training and validation accuracy steadily increasing. The validation accuracy reaches its peak without any noticeable overfitting, indicating the model has learned effectively.

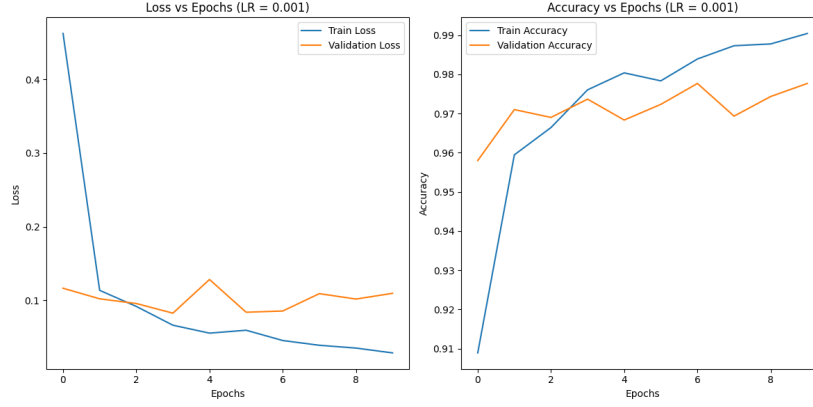


Figure 3: Loss and Accuracy vs Epochs for Unfreeze Last Block + Classifier (LR = 0.001)

Observations for Figure 3:

- **Loss and Accuracy Plot (LR = 0.001, Unfreeze Last Block):** The model's training accuracy increases rapidly, and the validation accuracy rises sharply, reaching **97.77%**, the highest among all strategies. There is no overfitting, and both training and validation losses decrease steadily.

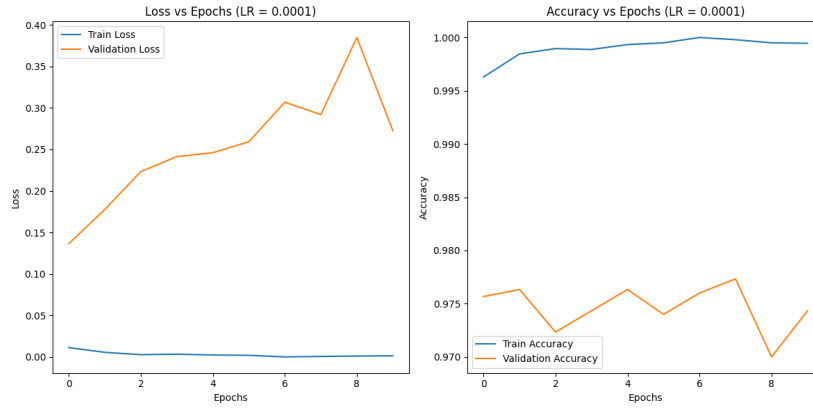


Figure 4: Loss and Accuracy vs Epochs for Unfreeze Last Block + Classifier (LR = 0.0001)

Observations for Figure 4:

- **Loss and Accuracy Plot (LR = 0.0001, Unfreeze Last Block):** The model initially shows good progress, but the training accuracy reaches an impressive **99.95%**, and validation accuracy stabilizes around 97.43%.

Overfitting starts to appear after epoch 5, reflected in the rising validation loss while the training loss continues to decrease.

5 Test Performance

After training, we evaluated the final model on the test set. The accuracy on the test set is reported below:

$$\text{Final Test Accuracy} = 97.93\%$$

This result demonstrates that the model generalizes well to unseen data, achieving strong performance on the test set.

6 Discussion

Based on the training and validation performance, the best configuration for fine-tuning was:

- **Unfreezing the last convolutional block and classifier head**
- **Learning rate = 0.001**

This setup provided the highest validation accuracy and demonstrated the ability to generalize effectively to new data. Lower learning rates, such as 0.0001, resulted in improved stability during training but showed slight overfitting towards the later epochs.

The final test accuracy of 97.93% confirms the model's strong generalization ability, particularly when the last block was fine-tuned.

7 Conclusion

In this experiment, we successfully applied fine-tuning techniques to a pre-trained VGG-16 model and compared two strategies. We concluded that unfreezing the last convolutional block, along with using a learning rate of 0.001, resulted in the best performance. This configuration achieved a final test accuracy of 97.93%, showcasing the power of transfer learning and fine-tuning in binary image classification tasks.